

Varun K S

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Summary

Python developer with hands-on experience building backend systems, REST APIs, and AI-integrated applications. Shipped production-ready Python solutions — an open-source DSA toolkit published on PyPI, an LLM-powered Flask platform, and a PostgreSQL-backed enterprise system. Proficient in Flask, FastAPI, OpenAI API integration, and agile SDLC workflows; proven collaborator in cross-functional engineering teams delivering scalable, data-driven digital products.

Technical Skills

Programming Languages: Python (OOP, Automation, Packaging), SQL, C/C++ (STL, OOP, Memory Management)
Backend & APIs: Flask, FastAPI, REST API Design, API Integration, API Authentication, Role-Based Access Control
AI Integration: OpenAI API, LLM Integration, Prompt Engineering, Sentiment Analysis
Data Processing: NumPy, Pandas, Perceptual Hashing
Databases: PostgreSQL, Database Design, Relational Schema Design, Normalization, DBMS
Testing & Quality: pytest, Unit Testing, Debugging, Code Review
Systems & Tools: Linux, Git/GitHub, Multithreading, TypeScript, Streamlit
Core CS: Data Structures & Algorithms, Operating Systems, System Design, Software Architecture, Agile/Spiral SDLC

Experience

Genesis Training and Consulting Service Pvt Ltd Feb 2026 – May 2026
Lead Software Engineering Intern | Bengaluru, India

- Led an **11-member cross-functional engineering team** in developing an AI-first digital recommendation platform — directing Python-based AI pipeline architecture, REST API layer design, conversational search feature planning, and AI-driven product discovery workflows aligned with modern digital transformation goals.
- Designed the platform’s **system architecture**, including infrastructure setup, deployment strategy, scalable repository organization, and folder structures enabling long-term maintainability across parallel workstreams.
- Established **Git/GitHub collaboration practices**, pull request review workflows, and engineering standards; managed Spiral SDLC sprint planning, milestone coordination, and structured release cycles for the full team.

Cognifyz Technologies Nov 2025 – Jan 2026
Software Development Intern

- Engineered **CrossBench**, a **C++** analytics engine using STL containers and OOP design patterns to consolidate AI benchmark data from **3 authoritative sources** into a unified cross-model performance leaderboard.
- Built automated **HTTP data ingestion pipelines** for structured retrieval, parsing, and normalization of live benchmark datasets — eliminating manual data collection overhead with memory-efficient, reproducible processing.

Projects

AlgoNest — Open-Source Python DSA Toolkit (PyPI) Python, Algorithms, Open Source, pytest

- Designed and published an open-source Python library on **PyPI** spanning **14 algorithm domains** — search, sort, graphs, dynamic programming, heaps, and greedy — with typed, iterable-first APIs and **50+ reusable primitives**; installable via `pip install algonest`.
- Structured with modular architecture, **pytest**-based test suites, public API documentation, and contribution guidelines — demonstrating end-to-end Python packaging and production-grade software engineering practices.

Smara AI — LLM-Powered Flask Platform Python, Flask, OpenAI API, REST API, Sentiment Analysis

- Built a **Flask REST API** backend integrating the **OpenAI API** for LLM-driven personalized insights, **AI-generated recommendations, and sentiment analysis** — with encrypted local storage, role-based authentication, and structured API endpoint design.

Campus Placement Management System React, TypeScript, PostgreSQL, REST API, RBAC

- Designed normalized **PostgreSQL** relational schemas with **RBAC** supporting concurrent multi-role workflows — students, recruiters, and coordinators — with real-time tracking, automated notifications, and analytics dashboards reducing coordinator overhead.

MediaCleaner — Python Media Deduplication Engine Python, OpenCV, Perceptual Hashing

- Built a **perceptual hashing**-based duplicate detection engine in Python using **OpenCV** for memory-efficient large-scale media deduplication; packaged as a cross-platform desktop application and commercially released on Gumroad.

Education

JSS Academy of Technical Education — Bengaluru Oct 2022 – Jul 2026 (Expected)
B.E. Computer Science and Engineering

Certifications

100 Days of Code: Python Pro Bootcamp — Udemy	Mar 2025
Meta Database Engineer Professional Certificate — Coursera	Mar 2025
Mastering Data Structures & Algorithms in C/C++ — Udemy	Mar 2024